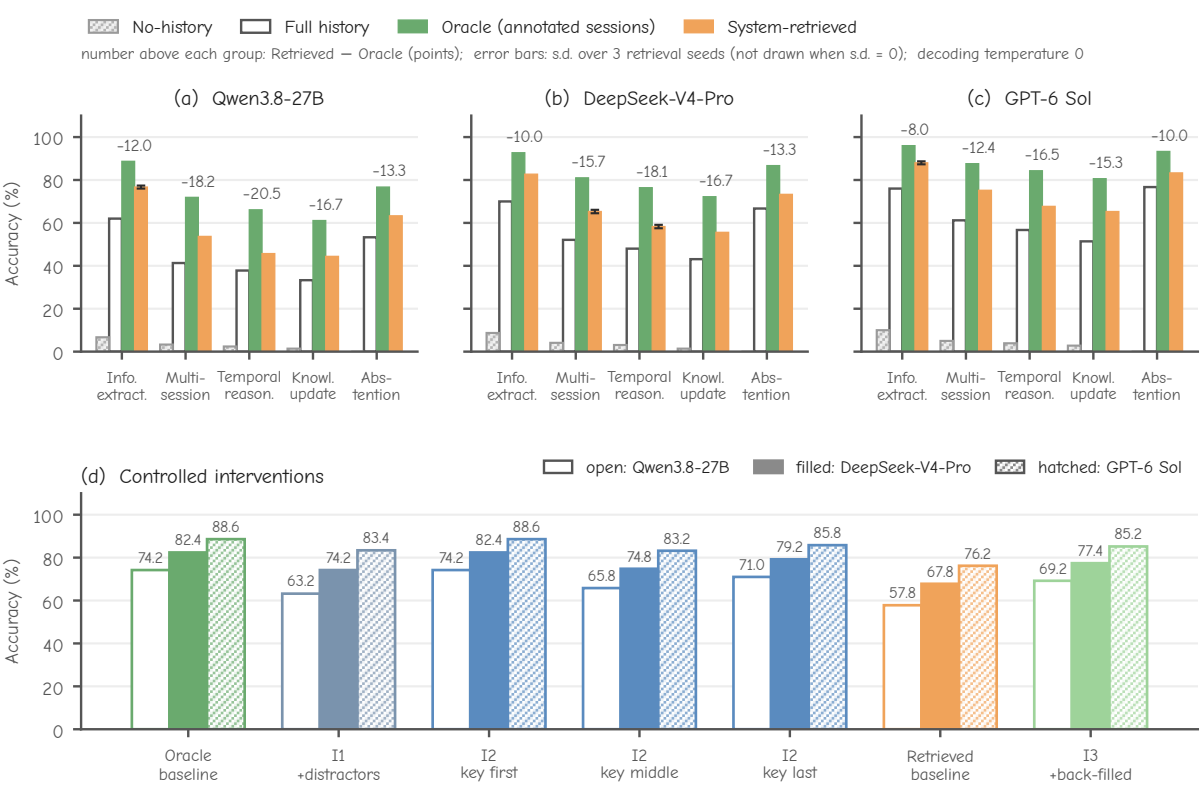

No-history
Full history
Oracle (annotated sessions)
System-retrieved
number above each group: Retrieved − Oracle (points); error bars: s.d. over 3 retrieval seeds (not drawn when s.d. = 0); decoding temperature 0
(a) Qwen3.8-27B
(b) DeepSeek-V4-Pro
(c) GPT-6 Sol
Accuracy (%)
-12.0
-18.2
-20.5
-16.7
-13.3
-10.0
-15.7
-18.1
-16.7
-13.3
-8.0
-12.4
-16.5
-15.3
-10.0
Info. extract.
Multi-session
Temporal reason.
Knowl. update
Abs-tention
(d) Controlled interventions
open: Qwen3.8-27B
filled: DeepSeek-V4-Pro
hatched: GPT-6 Sol
74.2
82.4
88.6
63.2
74.2
83.4
74.2
82.4
88.6
65.8
74.8
83.2
71.0
79.2
85.8
57.8
67.8
76.2
69.2
77.4
85.2
Oracle baseline
I1 +distractors
I2 key first
I2 key middle
I2 key last
Retrieved baseline
I3 +back-filled